

Euclid's Elements — Book I

Proposition 25

Formal Reconstruction — Technical Documentation

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If two triangles have two sides equal to two sides respectively, but have the base greater than the base, they will also have the one of the angles contained by the equal straight lines greater than the other.

1 Introduction

Euclid's Proposition I.25 is the converse comparison theorem to Proposition I.24.

Suppose two triangles have two corresponding sides congruent:

$$AB \cong DE, \quad AC \cong DF.$$

If the third side of the first triangle is greater than the third side of the second,

$$EF < BC,$$

then the included angle of the first triangle is greater:

$$\angle EDF < \angle BAC.$$

The formal reconstruction is strikingly short. Unlike Proposition I.24, no auxiliary geometric construction is needed inside the proposition itself. The proof is an elimination argument based on the previously established trichotomy for strict angle comparison.

The difficult geometric work needed for I.25 has already been absorbed into the Hilbert interface, in particular into `angle_trichotomy`. Once that theorem is available, Proposition I.25 becomes almost purely logical.

2 The Classical Configuration

Let $\triangle ABC$ and $\triangle DEF$ satisfy

$$AB \cong DE, \quad AC \cong DF,$$

and suppose

$$EF < BC.$$

The conclusion is

$$\angle EDF < \angle BAC.$$

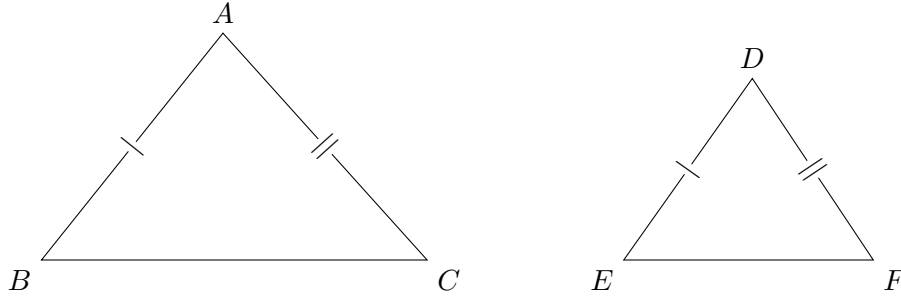


Figure 1: The configuration of Proposition I.25.

No new point is constructed in Proposition I.25. The same two triangles are compared throughout the proof.

3 Classical Proof

Euclid excludes the two alternatives incompatible with the required conclusion.

First suppose

$$\angle BAC \cong \angle EDF.$$

Then Proposition I.4 (SAS), applied to the two pairs of congruent sides, gives

$$BC \cong EF,$$

contradicting $EF < BC$.

Next suppose

$$\angle BAC < \angle EDF.$$

Apply Proposition I.24 with the two triangles interchanged. Since the included angle of $\triangle DEF$ is greater, I.24 gives

$$BC < EF,$$

again contradicting $EF < BC$.

Therefore the only remaining possibility is

$$\angle EDF < \angle BAC.$$

4 Angle Trichotomy

The decisive interface result is `angle_trichotomy`. For two nondegenerate angles it provides exactly the three alternatives

$$\angle BAC \cong \angle EDF, \quad \angle BAC < \angle EDF, \quad \angle EDF < \angle BAC.$$

This theorem was established in the Hilbert layer immediately before the formalization of I.25. Its proof uses Hilbert angle construction, same-side information, the order of rays in a half-plane, and the principle that two collinear points lying on the same side of a line through the vertex determine the same ray.

Thus angle trichotomy is not an additional axiom introduced for I.25. It is a derived structural theorem about the already defined relation `HilbertAngleLess`.

5 Proof Architecture of the Reconstruction

The formal proof is a three-branch elimination argument. Its complete architecture is short enough to display directly in ASCII:

```
angle_trichotomy
|
+-- angles congruent
|   -> SAS
|   -> BC ~= EF
|   -> contradiction with EF < BC
|
+-- BAC < EDF
|   -> Proposition I.24
|   -> BC < EF
|   -> contradiction with EF < BC
|
+-- EDF < BAC
    -> goal
```

This is not merely a summary of the Lean tactics. It is the actual mathematical dependency structure of the reconstructed proof.

The first branch is eliminated by triangle congruence. The second is eliminated by the direct hinge theorem I.24. The third branch is already the desired statement.

Consequently, after angle trichotomy has been established, I.25 contains no hidden construction and no additional geometric case analysis.

6 Hilbert Reconstruction

6.1 The Congruent-Angle Branch

Assume

$$\angle BAC \cong \angle EDF.$$

Together with

$$AB \cong DE, \quad AC \cong DF,$$

Hilbert SAS yields

$$BC \cong EF.$$

The interface theorem `hilbert_segmentLess_not_congruent` states that a strictly shorter segment cannot be congruent to the longer one. Hence this branch is impossible.

6.2 The Wrong Strict-Inequality Branch

Assume

$$\angle BAC < \angle EDF.$$

Interchange the roles of the triangles and reverse the side congruences:

$$DE \cong AB, \quad DF \cong AC.$$

Then Proposition I.24 gives

$$BC < EF.$$

But the original hypothesis is

$$EF < BC.$$

The asymmetry theorem `hilbert_segmentLess_asymm` excludes these simultaneous strict inequalities.

6.3 The Desired Branch

The remaining case is exactly

$$\angle EDF < \angle BAC,$$

so no further argument is needed.

7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_25
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : not (PrimCollinear Geo A B C))
  (hDEF : not (PrimCollinear Geo D E F))
  (hAB_DE : Geo.Congruent A B D E)
  (hAC_DF : Geo.Congruent A C D F)
  (hEF_BC : HilbertSegmentLess Geo E F B C) :
  HilbertAngleLess Geo E D F B A C
```

The proof begins by invoking `angle_trichotomy` on the two included angles.

In the equality branch, `SAS` produces a `TriangleCongruenceResult`; its field `sideBC` gives $BC \cong EF$. After reversing the congruence, the contradiction is discharged by `hilbert_segmentLess_not_congruent`.

In the second branch, Proposition I.24 is called with the two triangles interchanged. This gives $BC < EF$, contradicting the hypothesis through `hilbert_segmentLess_asymm`.

The third branch is returned directly.

8 Relationship with Earlier Propositions

8.1 Proposition I.4

Proposition I.4 supplies the SAS criterion used to eliminate the equality branch:

$$\angle BAC \cong \angle EDF, \quad AB \cong DE, \quad AC \cong DF \quad \implies \quad BC \cong EF.$$

8.2 Proposition I.24

Proposition I.24 is the central direct dependency of I.25. I.24 states that, under two pairs of congruent sides, a greater included angle implies a greater third side. I.25 reverses the comparison: a greater third side implies a greater included angle.

The formal proof of I.25 uses I.24 exactly once, to eliminate the wrong direction of angle inequality.

Together they give, under

$$AB \cong DE, \quad AC \cong DF,$$

the order correspondence

$$\angle EDF < \angle BAC \quad \iff \quad EF < BC.$$

9 Hilbert and Book Zero Components Used

9.1 `angle_trichotomy`

This is the principal structural result exposed by I.25. It converts the comparison of two nondegenerate angles into an exhaustive three-way split: congruent, smaller in one direction, or smaller in the other direction.

9.2 SAS

The existing SAS theorem converts equality of the included angles and the two side congruences into congruence of the third sides.

9.3 `hilbert_segmentLess_not_congruent`

This theorem expresses incompatibility between strict segment order and segment congruence. It eliminates the first trichotomy branch.

9.4 `hilbert_segmentLess_asymm`

This theorem expresses asymmetry of strict segment comparison:

$$AB < CD \quad \implies \quad \neg(CD < AB).$$

It eliminates the second branch after I.24 produces the opposite inequality.

10 Formalization Notes

Proposition I.25 provides a useful contrast with I.24.

I.24 required substantial geometric work. Its formal reconstruction had to expose point-order cases hidden by Euclid’s diagram and develop the corresponding hinge argument in each configuration.

I.25 does not repeat any of that geometry. Instead, the reconstruction first strengthens the reusable Hilbert interface by proving angle trichotomy. Once this has been done, I.25 is a direct logical consequence of

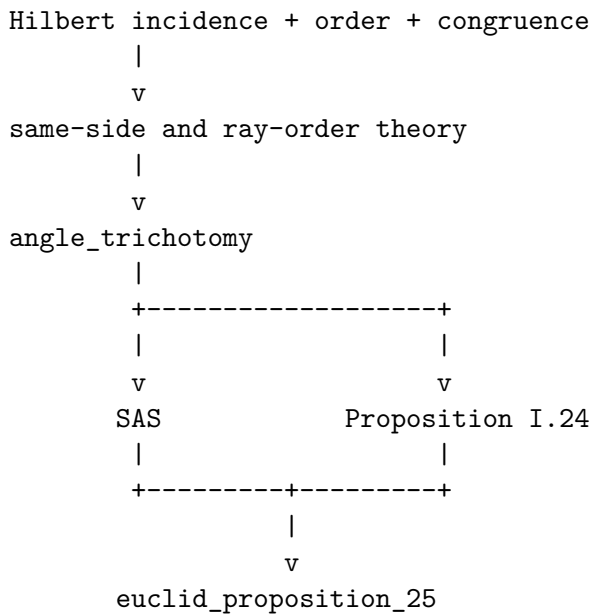
angle trichotomy, SAS, I.24.

This separation is important for the architecture of the project. A long foundational or order-theoretic argument should not be buried inside every later Euclidean proposition that needs it. Once identified, it becomes part of the intermediate synthetic language and later proofs become correspondingly transparent.

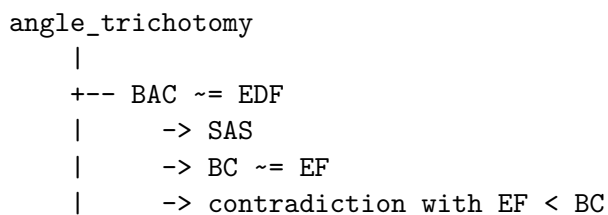
I.25 is therefore a particularly clean example of the role of the Hilbert interface: difficult geometric infrastructure is proved once, after which a classical theorem can recover its natural short proof.

11 Dependency Summary

The underlying dependency graph is



At the level of the final theorem:



```

|
+--- BAC < EDF
|     -> I.24
|     -> BC < EF
|     -> contradiction with EF < BC
|
+--- EDF < BAC
      -> exact

```

No new geometric axiom, continuity principle, coordinate system, or numerical angle measure is introduced.

12 Conclusion

Proposition I.25 is the converse comparison theorem to Proposition I.24.

Its formal reconstruction is exceptionally concise, but the concision is not accidental. The theorem becomes short only after the underlying angle-order theory has been completed.

The proof begins with angle trichotomy. Congruence of the included angles is excluded by SAS, because it would force congruence of the third sides. The wrong strict inequality is excluded by Proposition I.24, because it would force the opposite strict comparison of the third sides. The remaining branch is precisely the desired conclusion.

Thus I.25 gives a particularly clear instance of the architecture

foundational Hilbert theory $\longrightarrow$ reusable synthetic interface $\longrightarrow$ short Euclidean theorem.